Lab 1 - Code and Outputs

lab1_compare.py

```
"""Lab 1 - Section 1.5: symbolic rule vs probability threshold.
NOTE: probabilities are invented demonstration inputs. No model is trained,
so this is a BEHAVIOUR comparison, not an accuracy experiment.
Run: python lab1_compare.py
"""


def symbolic(score, complete):
    return "Review" if complete and score >= 70 else "Hold"


def threshold(probability):
    return "Review" if probability >= 0.70 else "Hold"


cases = [
    ("complete_high", 82, True, 0.81),
    ("complete_low", 68, True, 0.74),
    ("incomplete_high", 91, False, 0.88),
    ("boundary", 70, True, 0.70),
]

print("case | symbolic | threshold | disagreement")
for name, score, complete, probability in cases:
    a, b = symbolic(score, complete), threshold(probability)
    print(name, a, b, a != b, sep=" | ")

# Section 1.6 isolation test: change boundary score and probability separately
print("\n--- Section 1.6: isolating the boundary case ---")
print("score=69, complete=True        ->", symbolic(69, True))
print("probability=0.69               ->", threshold(0.69))
print("score=70, complete=False       ->", symbolic(70, False))
```

lab1_rules.py

```
"""Lab 1 - Task 2: rule-based scholarship screening with two extra conditions.
All thresholds are SYNTHETIC LAB ASSUMPTIONS, not real policy.
Review = flag for a human reviewer, NOT an award decision.
Run: python lab1_rules.py
"""

MIN_SCORE = 70          # given in the lab
```

```
MIN_GPA = 2.5           # new condition 1: min_gpa_met (synthetic)
LETTER_REQUIRED = True  # new condition 2:
recommendation_letter_present (synthetic)


def screen(score, documents_complete, gpa, has_letter):
    """Return (decision, reason) using only explicit, named
conditions."""
    if not documents_complete:
        return "Hold", "documents_complete failed"
    if score < MIN_SCORE:
        return "Hold", f"score {score} < {MIN_SCORE}"
    if gpa < MIN_GPA:
        return "Hold", f"min_gpa_met failed ({gpa} < {MIN_GPA})"
    if LETTER_REQUIRED and not has_letter:
        return "Hold", "recommendation_letter_present failed"
    return "Review", "all conditions satisfied"


# Exactly four core tests: pass, fail of each new condition, threshold
boundary.
tests = [
    # name,               score, docs_ok, gpa, letter, expected
    ("pass_all",            82, True, 3.2, True,  "Review"),
    ("fail_low_gpa",        85, True, 2.1, True,  "Hold"),
    ("fail_no_letter",      88, True, 3.5, False, "Hold"),
    ("boundary_exact",      70, True, 2.5, True,  "Review"),
]

print("case | input | expected | actual | reason | match")
all_ok = True
for name, score, docs, gpa, letter, expected in tests:
    actual, reason = screen(score, docs, gpa, letter)
    ok = actual == expected
    all_ok &= ok
    inp = f"score={score},docs={docs},gpa={gpa},letter={letter}"
    print(name, inp, expected, actual, reason, "PASS" if ok else
"FAIL", sep=" | ")

print("\nAll tests passed:", all_ok)
```

Output: compare_output.txt

```
case | symbolic | threshold | disagreement
complete_high | Review | Review | False
complete_low | Hold | Review | True
incomplete_high | Hold | Review | True
boundary | Review | Review | False

--- Section 1.6: isolating the boundary case ---
```

```
score=69, complete=True        -> Hold
probability=0.69               -> Hold
score=70, complete=False       -> Hold
```

Output: rules_output.txt

```
case | input | expected | actual | reason | match
pass_all | score=82,docs=True,gpa=3.2,letter=True | Review | Review |
all conditions satisfied | PASS
fail_low_gpa | score=85,docs=True,gpa=2.1,letter=True | Hold | Hold |
min_gpa_met failed (2.1 < 2.5) | PASS
fail_no_letter | score=88,docs=True,gpa=3.5,letter=False | Hold | Hold
| recommendation_letter_present failed | PASS
boundary_exact | score=70,docs=True,gpa=2.5,letter=True | Review |
Review | all conditions satisfied | PASS

All tests passed: True
```

README.md

Lab 1 – Introduction to AI & Paradigms (CAI601400)

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Contents

File	Purpose
`lab1_compare.py`	Symbolic rule vs probability threshold (Section 1.5 / 1.6)
`lab1_rules.py`	Rule system with two extra conditions + four core tests (Task 2)
`compare_output.txt`	Captured output of `lab1_compare.py`
`rules_output.txt`	Captured output of `lab1_rules.py`
`Lab1_Report.docx`	Full report (timeline, tests, disagreement table, comparison)

Setup

Requires Python 3.8+. No external packages.

```
python --version
```

Run

```
python lab1_compare.py
python lab1_rules.py
```

Assumptions

- All thresholds (score >= 70, GPA >= 2.5, letter required) are **synthetic lab assumptions**.
- Probabilities in `lab1_compare.py` are **invented demo inputs**; no model is trained.
- `Review` means “flag for a human reviewer”, not an award decision.
- No credentials, private data or virtual environments are included.